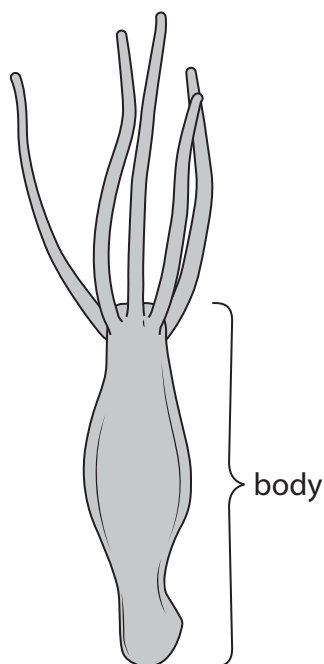


4 Hydra are small animals that live in water.

They have a cylindrical body.

The drawing shows a hydra.



(a) This hydra has a body length of 1.8 mm and a diameter of 1 mm.

(i) Which is the magnification of this drawing?

(1)

- ☐ **A** 3.75×10^{-2}
- ☐ **B** 3.75×10^{-1}
- ☐ **C** 2.67×10^0
- ☐ **D** 2.67×10^1

(ii) Which is the surface area, in mm^2 , of this hydra?

Use the formula:

$$\text{surface area} = 2\pi r l$$

(1)

- ☐ **A** 0.9π
- ☐ **B** 1.8π
- ☐ **C** 3.6π
- ☐ **D** 5.4π



- (iii) Another hydra has a surface area of $2.3\pi \text{ mm}^2$, a body length of 1.8 mm and a diameter of 1.2 mm.

Calculate the surface area to volume ratio of this hydra.

Use the formula:

$$\text{volume} = \pi r^2 l$$

(2)

Answer

- (b) A hydra has a hollow body that is made up of two layers of cells.

Explain how a hydra is adapted for gas exchange.

(3)

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(Total for Question 4 = 7 marks)

